

LTSpice Notch Quality Report

CROSS COUPLED4

Auto-detected swept parameters and quality-factor ($Q = f_0 / BW$ @ passband-3 dB) analysis

Run summary

Signal columns analyzed: 1

Total simulation steps : 36

V(act+,act-)
swept parameters: Cp, Itail

Report contents (per signal column):

- 1. All curves — Bode plot (magnitude + phase)
- 2. Magnitude grouped by each swept parameter (one subplot per fixed value)
- 3. Phase grouped by each swept parameter (one subplot per fixed value)
- 4. Quality-factor table for that column

End of report (extra analyses):

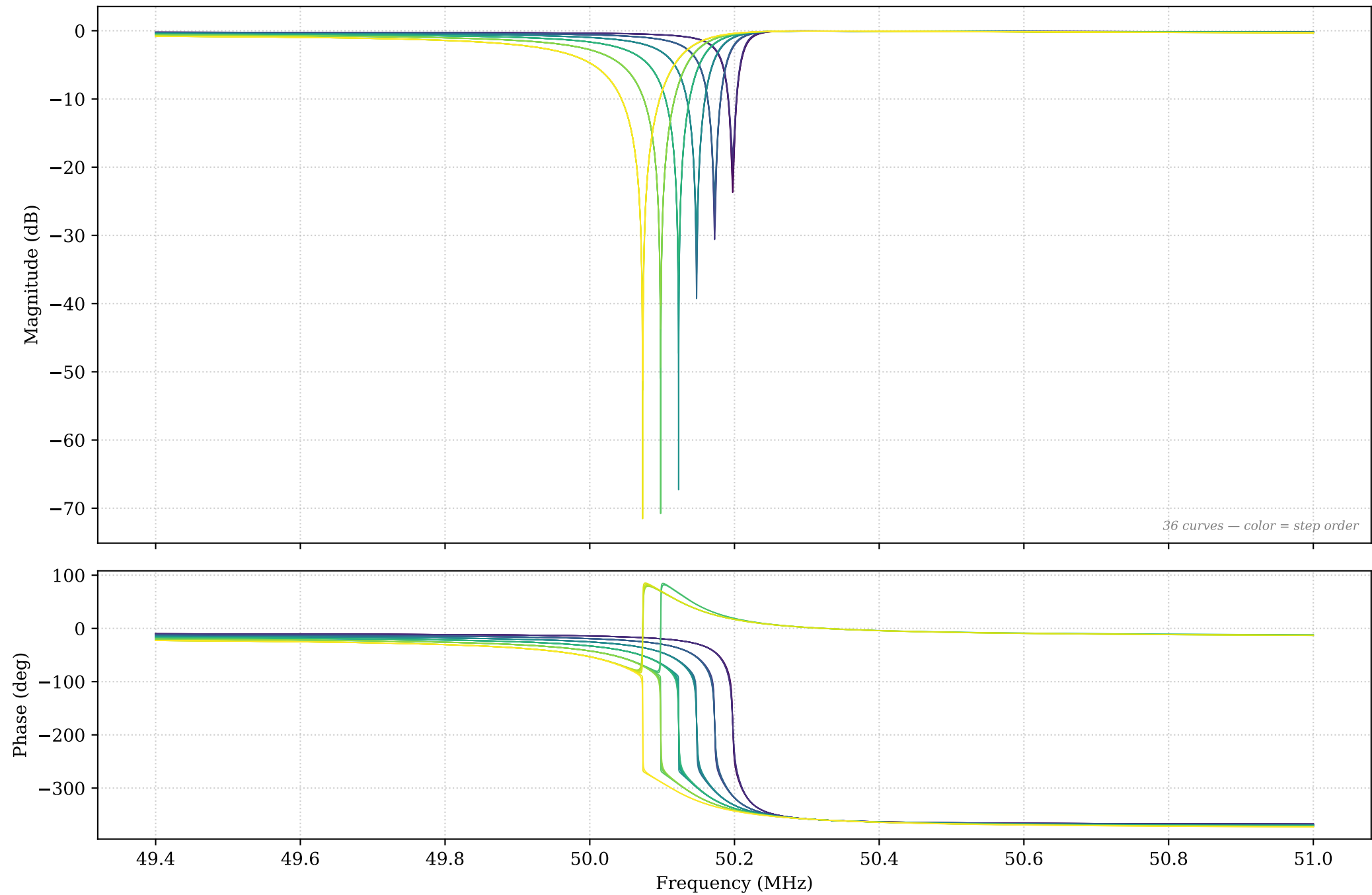
- A. Cross-signal comparison (act vs pas, if both present)
- B. Q vs each swept parameter (one line per other-param combo)
- C. BW vs each swept parameter
- D. Q heatmap for V(act): rows $\times$ columns swept params, colored
- E. Combined quality-factor table across all signals

Signal: $V(\text{act+}, \text{act-})$

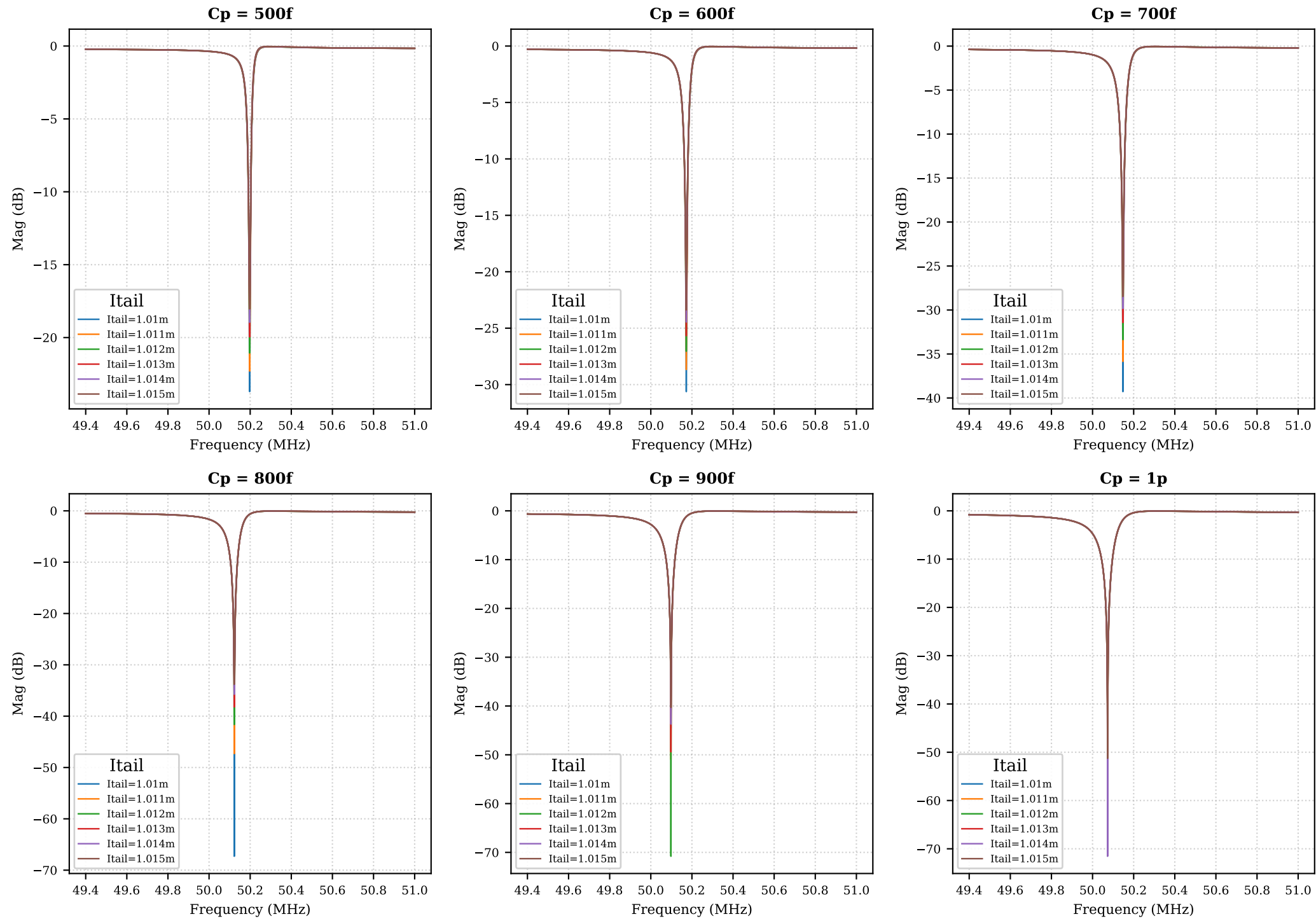
Swept parameters: C_p , I_{tail}

Steps: 36

V(act+,act-) — All curves (36 steps) — Bode

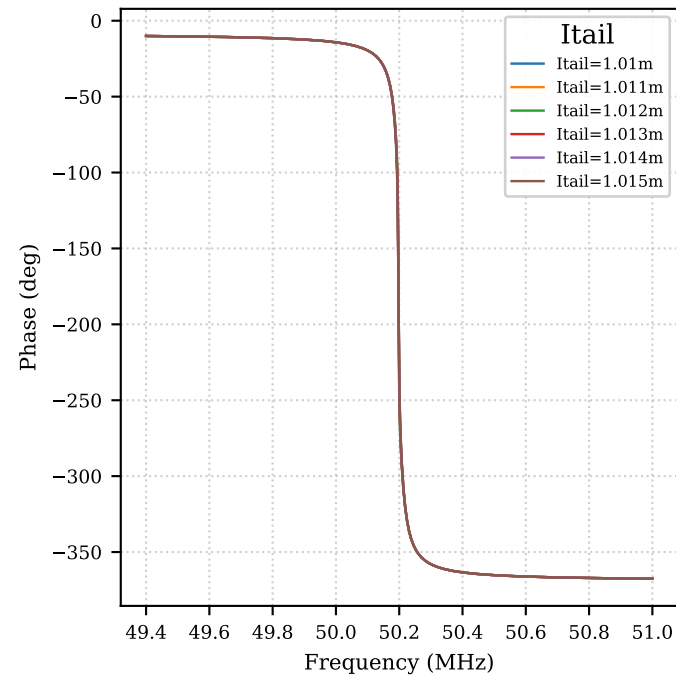


V(act+,act-) — Magnitude grouped by Cp (each subplot fixes Cp; legend = Itail)

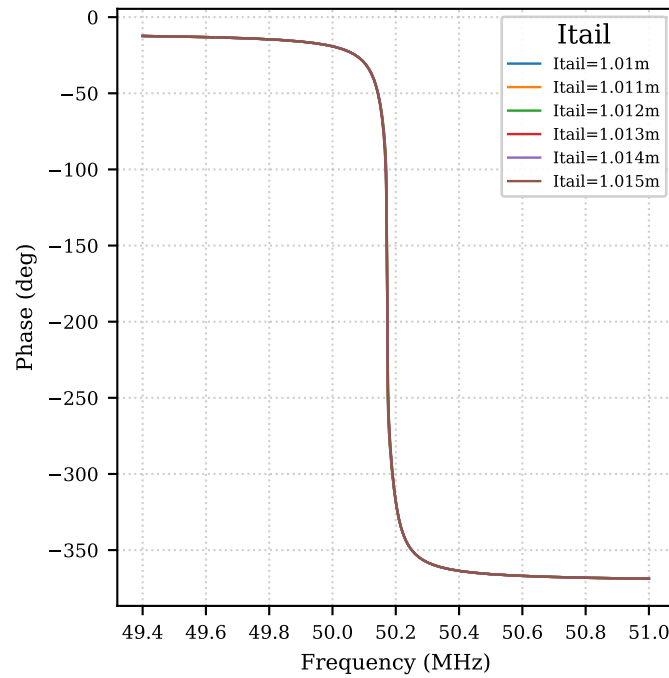


V(act+,act-) — Phase grouped by Cp (each subplot fixes Cp; legend = Itail)

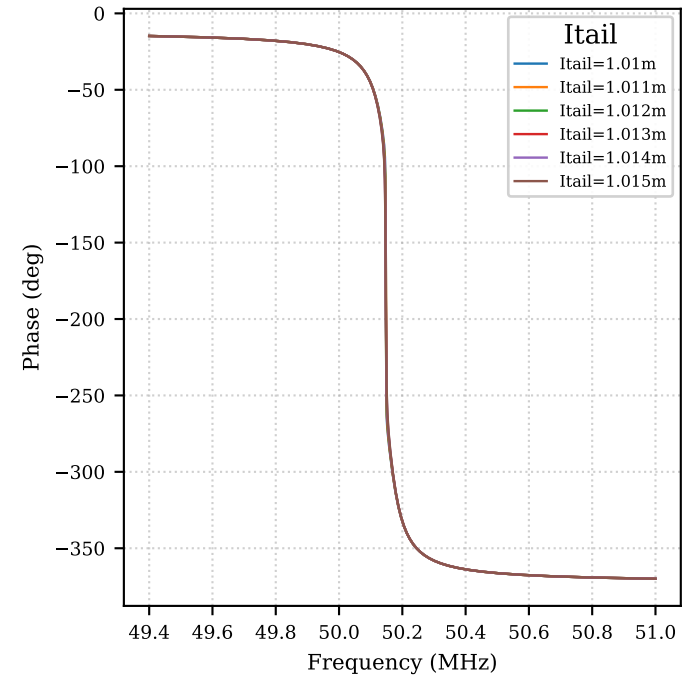
Cp = 500f



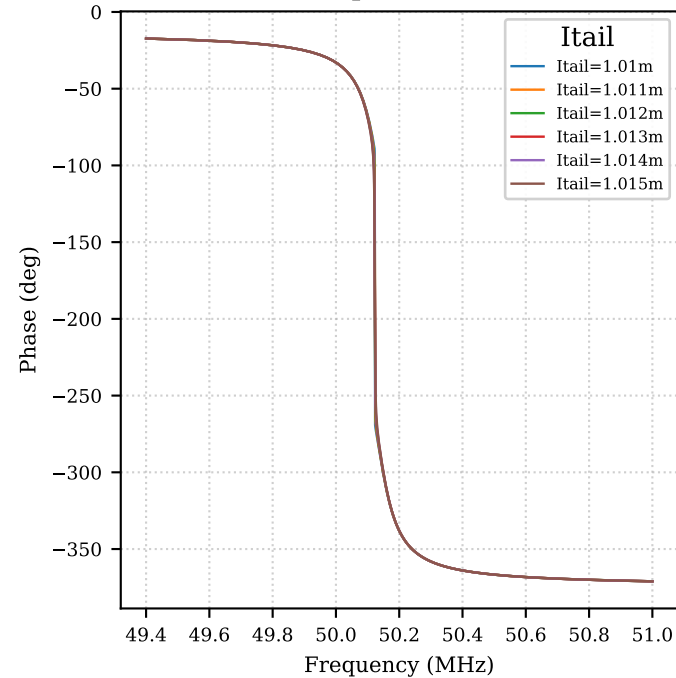
Cp = 600f



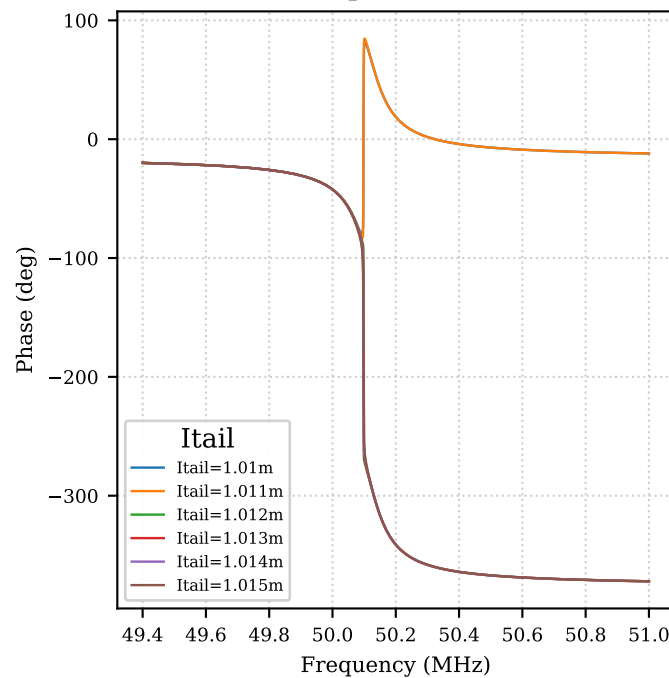
Cp = 700f



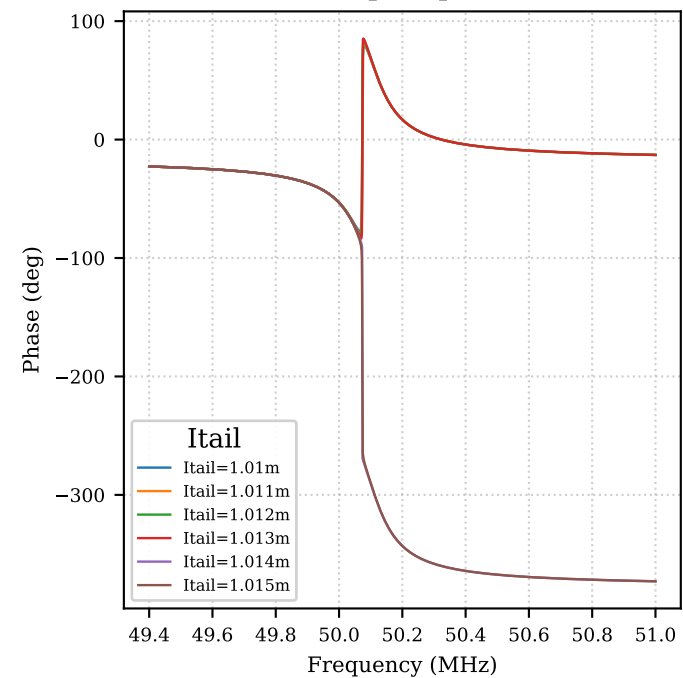
Cp = 800f



Cp = 900f

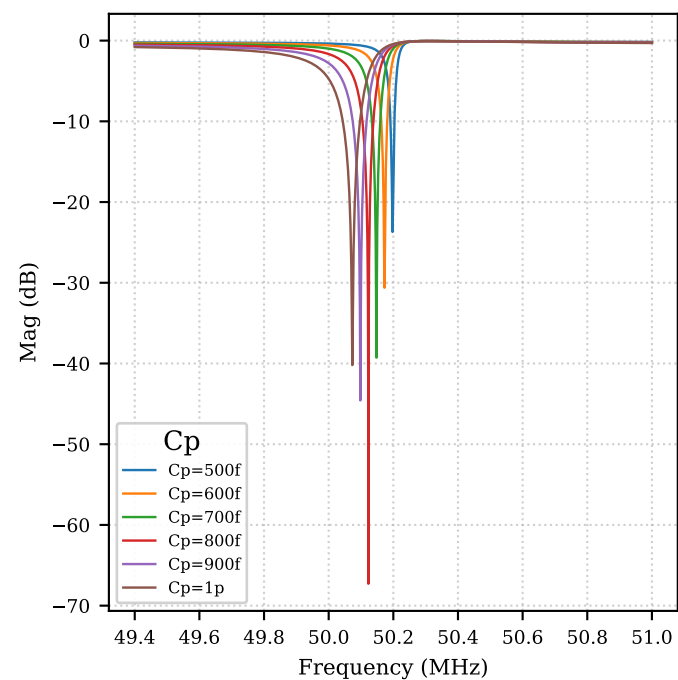


Cp = 1p

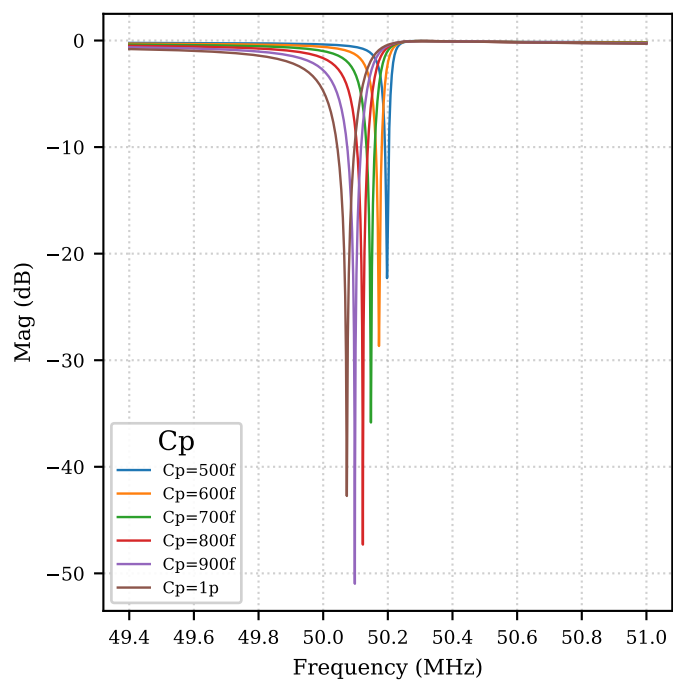


V(act+,act-) — Magnitude grouped by Itail (each subplot fixes Itail; legend = Cp)

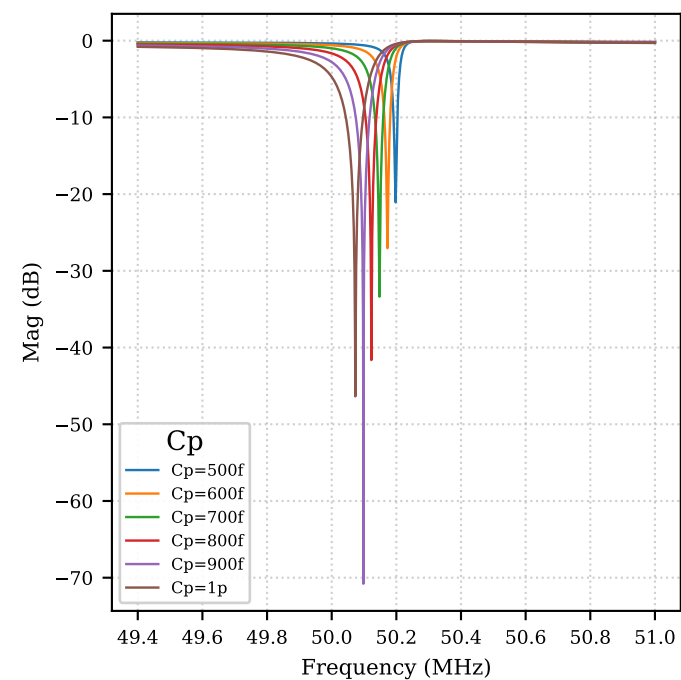
Itail = 1.01m



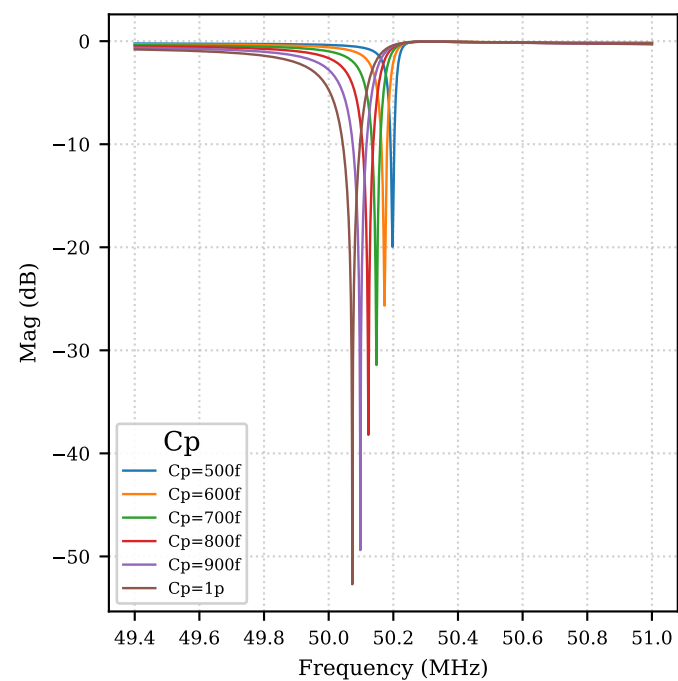
Itail = 1.011m



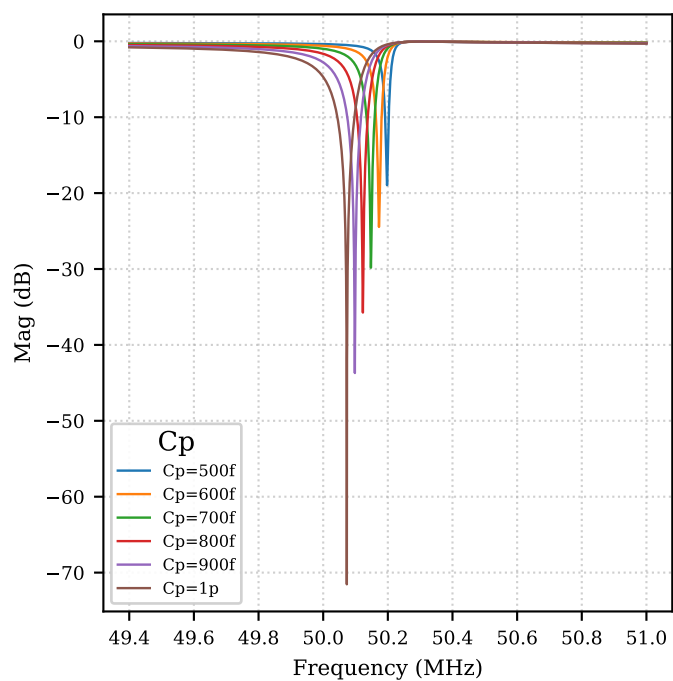
Itail = 1.012m



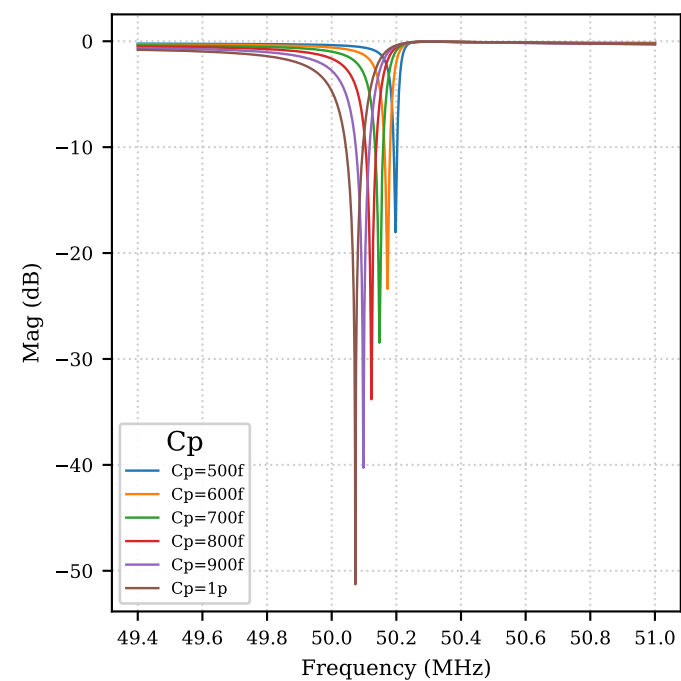
Itail = 1.013m



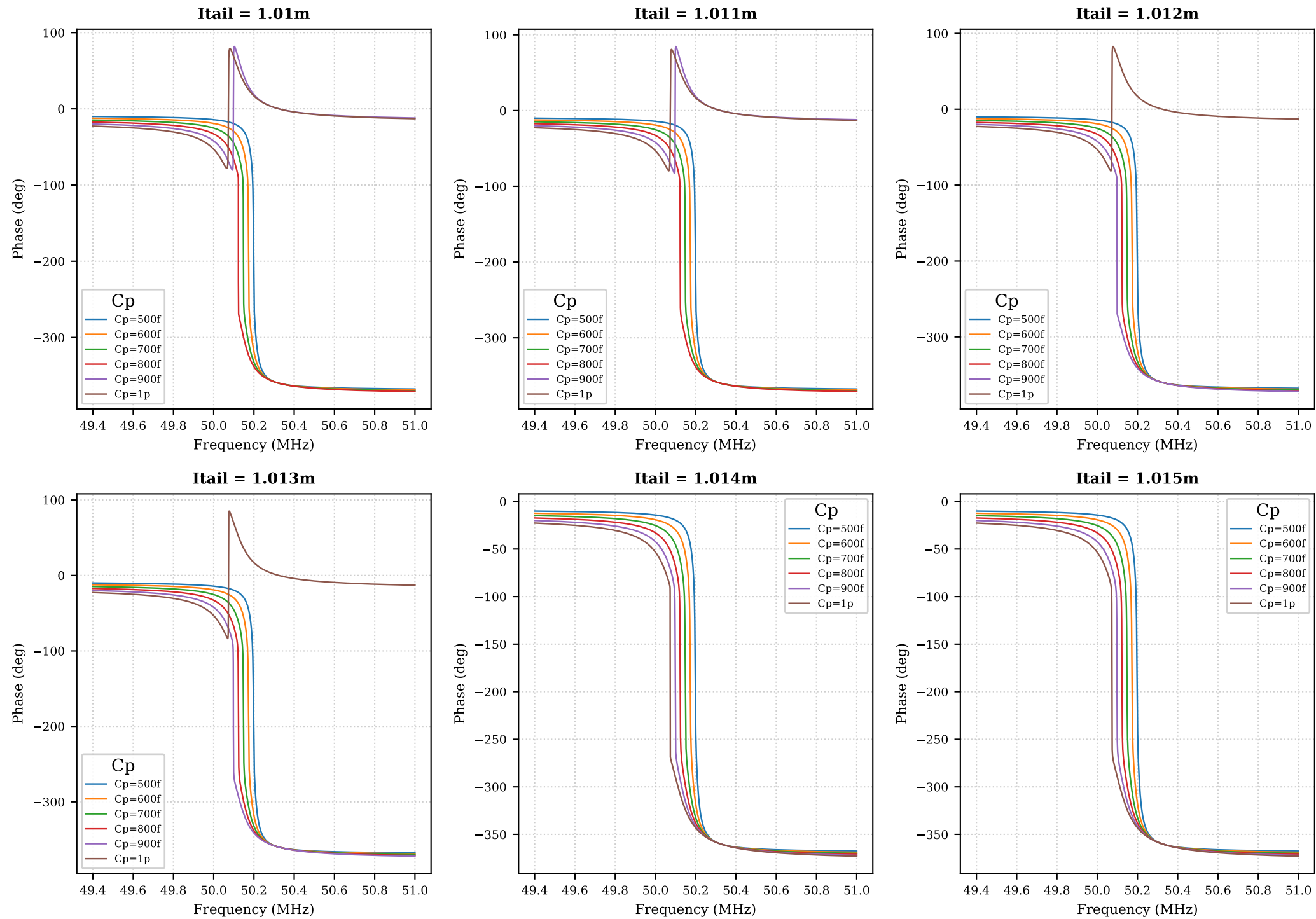
Itail = 1.014m



Itail = 1.015m



V(act+,act-) — Phase grouped by Itail (each subplot fixes Itail; legend = Cp)



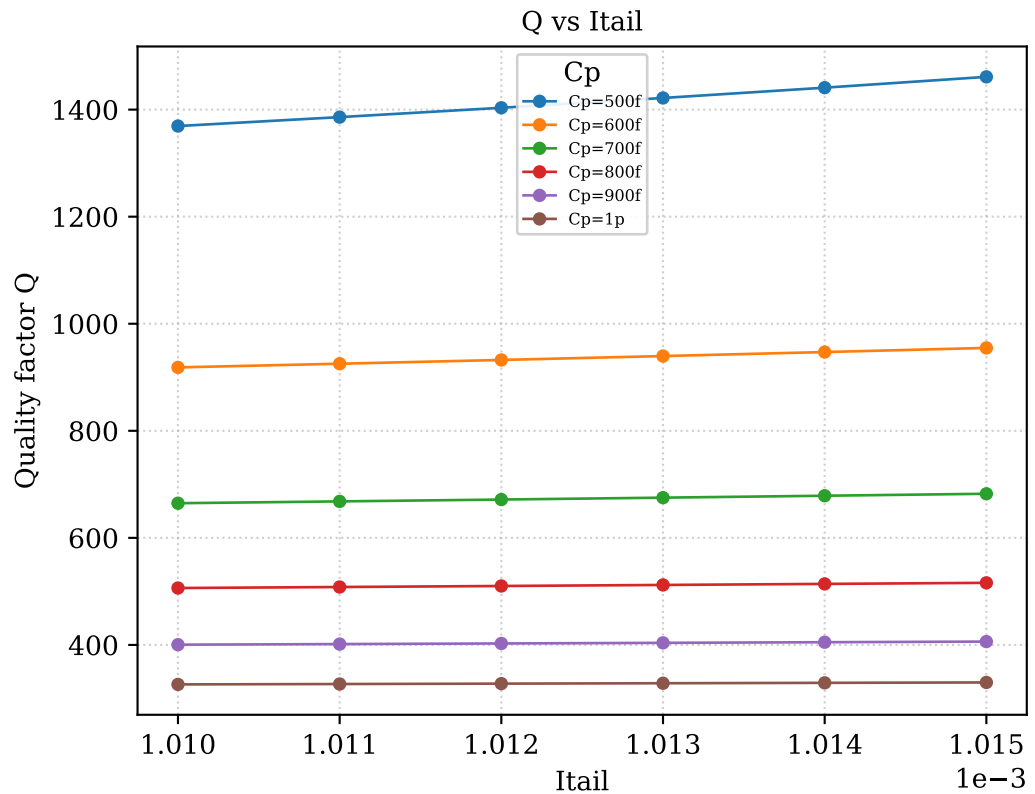
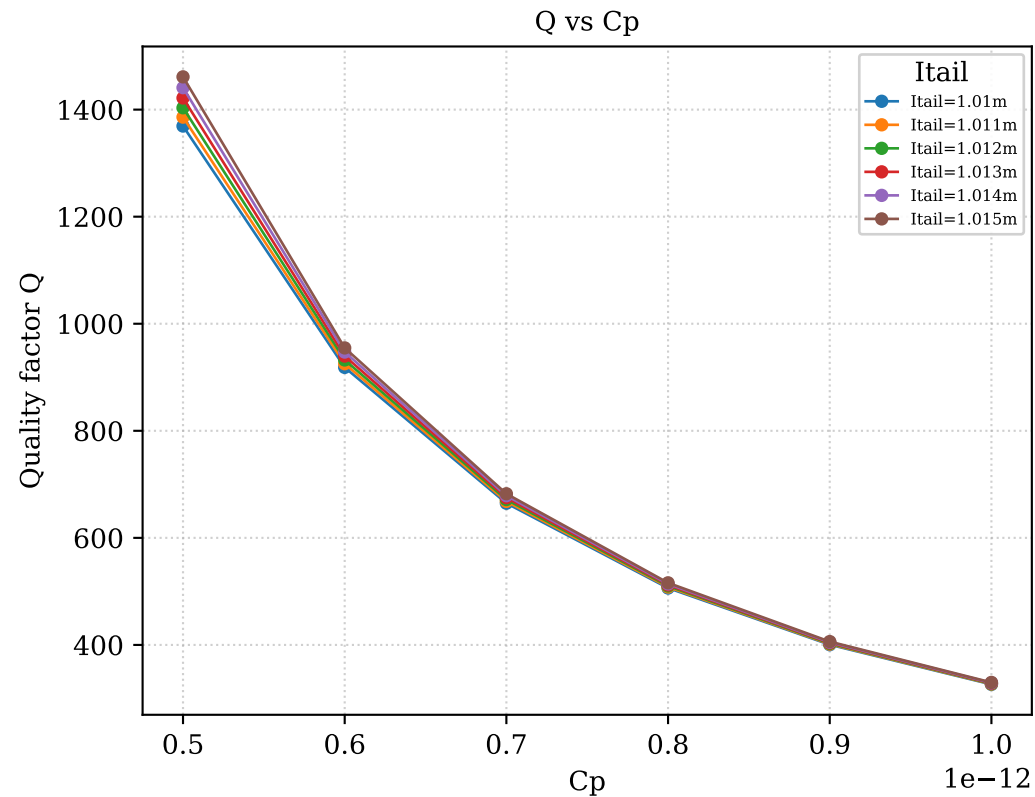
V(act+,act-) – Quality factor table (page 1/2)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	500f	1.01m	50.2	-0.1914	23.52	3.666e+04	1369	
V(act+,act-)	500f	1.011m	50.2	-0.1913	22.11	3.622e+04	1386	
V(act+,act-)	500f	1.012m	50.2	-0.1913	20.87	3.577e+04	1403	
V(act+,act-)	500f	1.013m	50.2	-0.1912	19.76	3.531e+04	1422	
V(act+,act-)	500f	1.014m	50.2	-0.1912	18.76	3.484e+04	1441	
V(act+,act-)	500f	1.015m	50.2	-0.1911	17.85	3.435e+04	1461	
V(act+,act-)	600f	1.01m	50.17	-0.2397	30.38	5.463e+04	918.5	
V(act+,act-)	600f	1.011m	50.17	-0.2396	28.42	5.422e+04	925.3	
V(act+,act-)	600f	1.012m	50.17	-0.2396	26.81	5.381e+04	932.4	
V(act+,act-)	600f	1.013m	50.17	-0.2395	25.43	5.34e+04	939.6	
V(act+,act-)	600f	1.014m	50.17	-0.2394	24.22	5.297e+04	947.1	
V(act+,act-)	600f	1.015m	50.17	-0.2393	23.14	5.255e+04	954.8	
V(act+,act-)	700f	1.01m	50.15	-0.2987	38.98	7.544e+04	664.7	
V(act+,act-)	700f	1.011m	50.15	-0.2986	35.54	7.506e+04	668.1	
V(act+,act-)	700f	1.012m	50.15	-0.2985	33.07	7.467e+04	671.6	
V(act+,act-)	700f	1.013m	50.15	-0.2984	31.13	7.428e+04	675.1	
V(act+,act-)	700f	1.014m	50.15	-0.2983	29.53	7.389e+04	678.7	
V(act+,act-)	700f	1.015m	50.15	-0.2982	28.17	7.349e+04	682.4	
V(act+,act-)	800f	1.01m	50.12	-0.3694	66.92	9.9e+04	506.3	
V(act+,act-)	800f	1.011m	50.12	-0.3693	46.93	9.863e+04	508.2	
V(act+,act-)	800f	1.012m	50.12	-0.3691	41.26	9.826e+04	510.1	
V(act+,act-)	800f	1.013m	50.12	-0.369	37.84	9.789e+04	512	
V(act+,act-)	800f	1.014m	50.12	-0.3689	35.37	9.751e+04	514	
V(act+,act-)	800f	1.015m	50.12	-0.3688	33.44	9.714e+04	516	
V(act+,act-)	900f	1.01m	50.1	-0.4528	44.12	1.251e+05	400.5	
V(act+,act-)	900f	1.011m	50.1	-0.4527	50.52	1.247e+05	401.6	
V(act+,act-)	900f	1.012m	50.1	-0.4525	70.34	1.244e+05	402.7	
V(act+,act-)	900f	1.013m	50.1	-0.4523	48.94	1.24e+05	403.9	
V(act+,act-)	900f	1.014m	50.1	-0.4522	43.25	1.237e+05	405.1	
V(act+,act-)	900f	1.015m	50.1	-0.452	39.83	1.233e+05	406.2	
V(act+,act-)	1p	1.01m	50.07	-0.5502	39.65	1.535e+05	326.2	
V(act+,act-)	1p	1.011m	50.07	-0.55	42.19	1.531e+05	327	

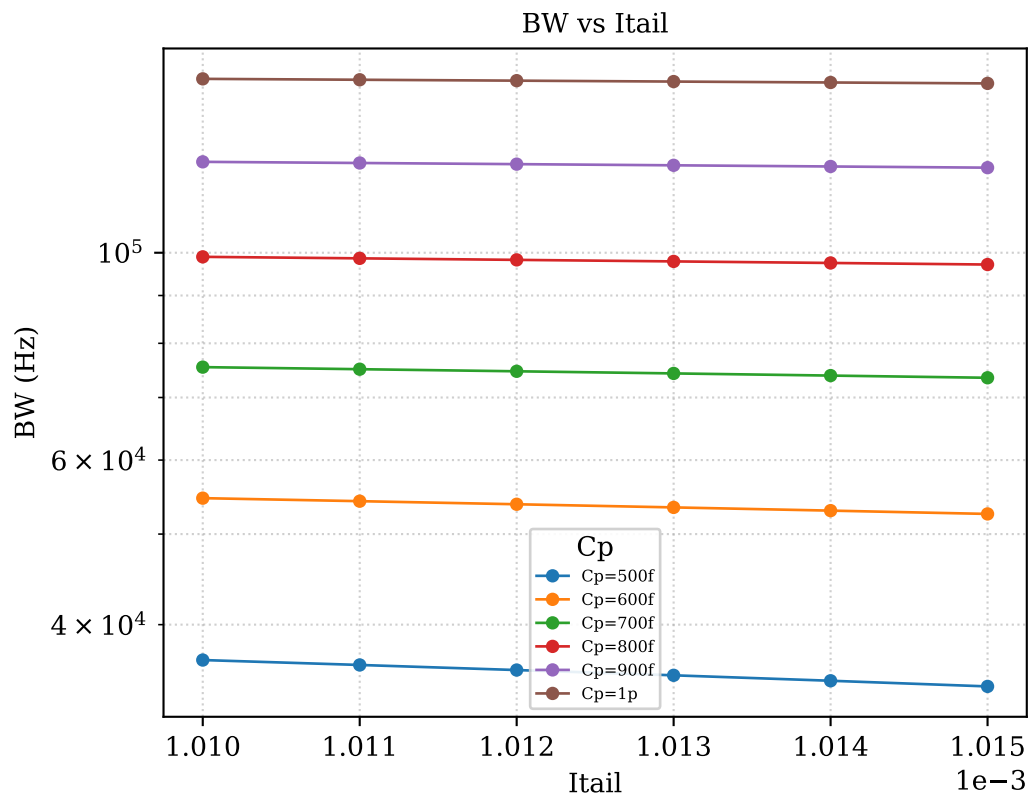
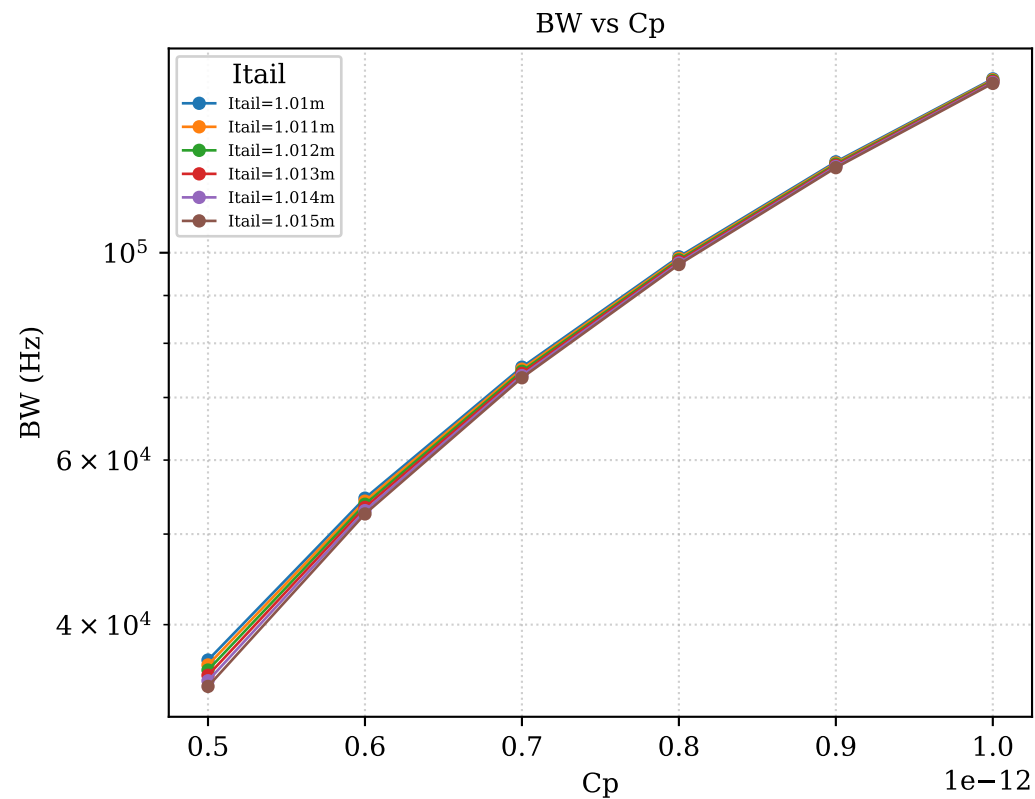
V(act+,act-) — Quality factor table (page 2/2)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	1p	1.012m	50.07	-0.5498	45.81	1.528e+05	327.7	
V(act+,act-)	1p	1.013m	50.07	-0.5496	52.16	1.525e+05	328.4	
V(act+,act-)	1p	1.014m	50.07	-0.5494	70.99	1.521e+05	329.2	
V(act+,act-)	1p	1.015m	50.07	-0.5492	50.73	1.518e+05	329.9	

B. V(act+,act-) — Quality factor vs swept parameters



C. V(act+,act-) — Bandwidth (-3 dB) vs swept parameters



D. V(act+,act-) — Q heatmap (rows = Cp, columns = Itail)

	Itail=1.01m	Itail=1.011m	Itail=1.012m	Itail=1.013m	Itail=1.014m	Itail=1.015m
Cp=500f	1369	1386	1403	1422	1441	1461
Cp=600f	918	925	932	940	947	955
Cp=700f	665	668	672	675	679	682
Cp=800f	506	508	510	512	514	516
Cp=900f	400	402	403	404	405	406
Cp=1p	326	327	328	328	329	330

Red = BW could not be measured | Yellow = Q computed ($Q \leq 1400$) | Green = $Q > 1400$

E. Combined quality-factor table – all signals (page 1/2)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	500f	1.01m	50.2	-0.1914	23.52	3.666e+04	1369	
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E. Combined quality-factor table — all signals (page 2/2)

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